

A Mutual Information of LMD

The MusicBERT OctupleMIDI token dictionary contains 256 bar tokens, 128 position tokens, 129 instrument tokens, 256 pitch tokens, 128 duration tokens, 32 velocity tokens, 254 time-signature tokens, and 49 tempo tokens. The panels show the unscaled plug-in MI (nats) between each of the eight attributes and the layer-wise cluster assignment on the held-out LMD-matched test split, for $K = 1,000$ (solid) and $K = 2,000$ (dashed), averaged over quantizer seeds (band: ± 1 SD). Per-seed values are in `tables/token_mi_lmd.csv`.

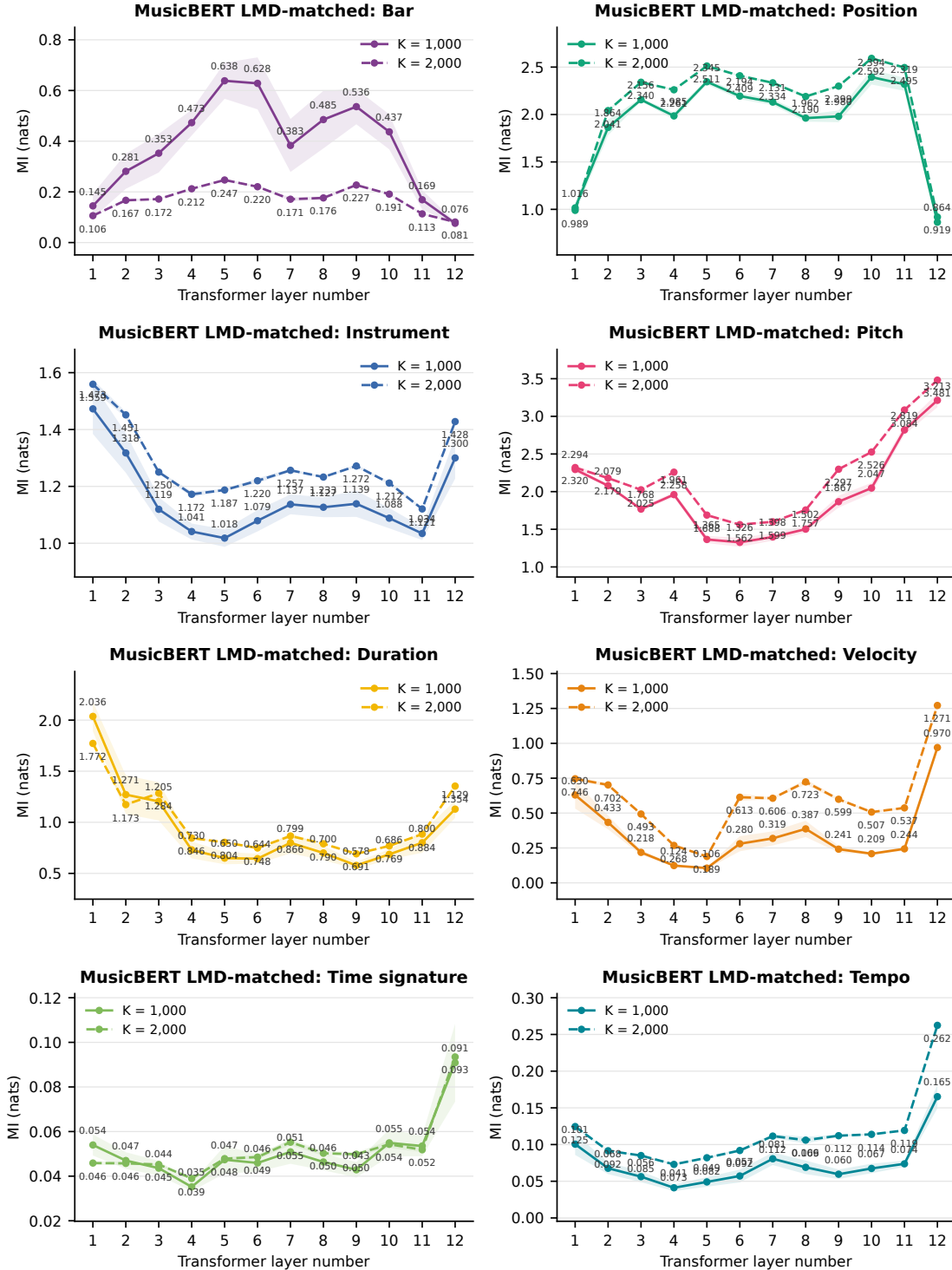


Table 1: LMD-matched: plug-in MI in nats at $K = 1,000$, mean $\pm$ SD over quantizer seeds (12, 42, 97); 4,472,244 note tokens.

Layer	Bar	Position	Instrument	Pitch	Duration	Velocity	Time signature	Tempo
1	0.145 $\pm$ 0.048	1.016 $\pm$ 0.052	1.473 $\pm$ 0.088	2.294 $\pm$ 0.037	2.036 $\pm$ 0.145	0.630 $\pm$ 0.098	0.054 $\pm$ 0.005	0.101 $\pm$ 0.015
2	0.281 $\pm$ 0.069	1.864 $\pm$ 0.084	1.318 $\pm$ 0.070	2.079 $\pm$ 0.015	1.271 $\pm$ 0.190	0.433 $\pm$ 0.050	0.047 $\pm$ 0.004	0.068 $\pm$ 0.008
3	0.353 $\pm$ 0.076	2.156 $\pm$ 0.017	1.119 $\pm$ 0.043	1.768 $\pm$ 0.020	1.205 $\pm$ 0.187	0.218 $\pm$ 0.018	0.044 $\pm$ 0.003	0.056 $\pm$ 0.007
4	0.473 $\pm$ 0.051	1.985 $\pm$ 0.038	1.041 $\pm$ 0.028	1.961 $\pm$ 0.035	0.730 $\pm$ 0.083	0.124 $\pm$ 0.005	0.035 $\pm$ 0.002	0.041 $\pm$ 0.005
5	0.638 $\pm$ 0.071	2.345 $\pm$ 0.029	1.018 $\pm$ 0.031	1.365 $\pm$ 0.047	0.650 $\pm$ 0.060	0.106 $\pm$ 0.007	0.047 $\pm$ 0.006	0.049 $\pm$ 0.007
6	0.628 $\pm$ 0.103	2.194 $\pm$ 0.034	1.079 $\pm$ 0.038	1.326 $\pm$ 0.060	0.644 $\pm$ 0.049	0.280 $\pm$ 0.054	0.046 $\pm$ 0.004	0.057 $\pm$ 0.009
7	0.383 $\pm$ 0.105	2.131 $\pm$ 0.032	1.137 $\pm$ 0.035	1.398 $\pm$ 0.049	0.799 $\pm$ 0.103	0.319 $\pm$ 0.051	0.051 $\pm$ 0.005	0.081 $\pm$ 0.009
8	0.485 $\pm$ 0.114	1.962 $\pm$ 0.038	1.127 $\pm$ 0.035	1.502 $\pm$ 0.046	0.700 $\pm$ 0.071	0.387 $\pm$ 0.059	0.046 $\pm$ 0.004	0.069 $\pm$ 0.009
9	0.536 $\pm$ 0.070	1.980 $\pm$ 0.062	1.139 $\pm$ 0.046	1.867 $\pm$ 0.079	0.578 $\pm$ 0.035	0.241 $\pm$ 0.024	0.043 $\pm$ 0.003	0.060 $\pm$ 0.007
10	0.437 $\pm$ 0.069	2.394 $\pm$ 0.078	1.088 $\pm$ 0.036	2.047 $\pm$ 0.070	0.686 $\pm$ 0.049	0.209 $\pm$ 0.008	0.055 $\pm$ 0.006	0.067 $\pm$ 0.006
11	0.169 $\pm$ 0.038	2.319 $\pm$ 0.069	1.034 $\pm$ 0.023	2.819 $\pm$ 0.061	0.800 $\pm$ 0.099	0.244 $\pm$ 0.014	0.054 $\pm$ 0.005	0.074 $\pm$ 0.004
12	0.076 $\pm$ 0.006	0.864 $\pm$ 0.027	1.300 $\pm$ 0.073	3.213 $\pm$ 0.093	1.129 $\pm$ 0.086	0.970 $\pm$ 0.022	0.091 $\pm$ 0.017	0.165 $\pm$ 0.016

Table 2: LMD-matched: plug-in MI in nats at $K = 2,000$, mean $\pm$ SD over quantizer seeds (1, 2, 3, 4, 42); 17,182,797 note tokens.

Layer	Bar	Position	Instrument	Pitch	Duration	Velocity	Time signature	Tempo
1	0.106 $\pm$ 0.001	0.989 $\pm$ 0.010	1.559 $\pm$ 0.017	2.320 $\pm$ 0.025	1.772 $\pm$ 0.015	0.746 $\pm$ 0.014	0.046 $\pm$ 0.001	0.125 $\pm$ 0.002
2	0.167 $\pm$ 0.006	2.041 $\pm$ 0.013	1.451 $\pm$ 0.011	2.179 $\pm$ 0.016	1.173 $\pm$ 0.015	0.702 $\pm$ 0.012	0.046 $\pm$ 0.001	0.092 $\pm$ 0.001
3	0.172 $\pm$ 0.002	2.340 $\pm$ 0.008	1.250 $\pm$ 0.004	2.025 $\pm$ 0.006	1.284 $\pm$ 0.006	0.493 $\pm$ 0.008	0.045 $\pm$ 0.000	0.085 $\pm$ 0.001
4	0.212 $\pm$ 0.002	2.261 $\pm$ 0.008	1.172 $\pm$ 0.010	2.258 $\pm$ 0.009	0.846 $\pm$ 0.013	0.268 $\pm$ 0.005	0.039 $\pm$ 0.001	0.073 $\pm$ 0.001
5	0.247 $\pm$ 0.006	2.511 $\pm$ 0.003	1.187 $\pm$ 0.004	1.688 $\pm$ 0.010	0.804 $\pm$ 0.006	0.189 $\pm$ 0.002	0.048 $\pm$ 0.001	0.082 $\pm$ 0.001
6	0.220 $\pm$ 0.003	2.409 $\pm$ 0.007	1.220 $\pm$ 0.009	1.562 $\pm$ 0.013	0.748 $\pm$ 0.008	0.613 $\pm$ 0.026	0.049 $\pm$ 0.001	0.092 $\pm$ 0.002
7	0.171 $\pm$ 0.007	2.334 $\pm$ 0.013	1.257 $\pm$ 0.005	1.599 $\pm$ 0.011	0.866 $\pm$ 0.010	0.606 $\pm$ 0.006	0.055 $\pm$ 0.001	0.112 $\pm$ 0.003
8	0.176 $\pm$ 0.007	2.190 $\pm$ 0.015	1.233 $\pm$ 0.003	1.757 $\pm$ 0.014	0.790 $\pm$ 0.004	0.723 $\pm$ 0.020	0.050 $\pm$ 0.001	0.106 $\pm$ 0.005
9	0.227 $\pm$ 0.003	2.299 $\pm$ 0.011	1.272 $\pm$ 0.005	2.297 $\pm$ 0.021	0.691 $\pm$ 0.005	0.599 $\pm$ 0.010	0.050 $\pm$ 0.001	0.112 $\pm$ 0.001
10	0.191 $\pm$ 0.004	2.592 $\pm$ 0.010	1.212 $\pm$ 0.005	2.526 $\pm$ 0.013	0.769 $\pm$ 0.007	0.507 $\pm$ 0.007	0.054 $\pm$ 0.002	0.114 $\pm$ 0.001
11	0.113 $\pm$ 0.003	2.495 $\pm$ 0.014	1.121 $\pm$ 0.005	3.084 $\pm$ 0.024	0.884 $\pm$ 0.011	0.537 $\pm$ 0.009	0.052 $\pm$ 0.001	0.119 $\pm$ 0.001
12	0.081 $\pm$ 0.001	0.919 $\pm$ 0.013	1.428 $\pm$ 0.008	3.481 $\pm$ 0.009	1.354 $\pm$ 0.019	1.271 $\pm$ 0.027	0.093 $\pm$ 0.003	0.262 $\pm$ 0.002

B Mutual Information of SOD

Same quantizers (fitted on the LMD-matched training split), evaluated on the Symbolic Orchestral Database.

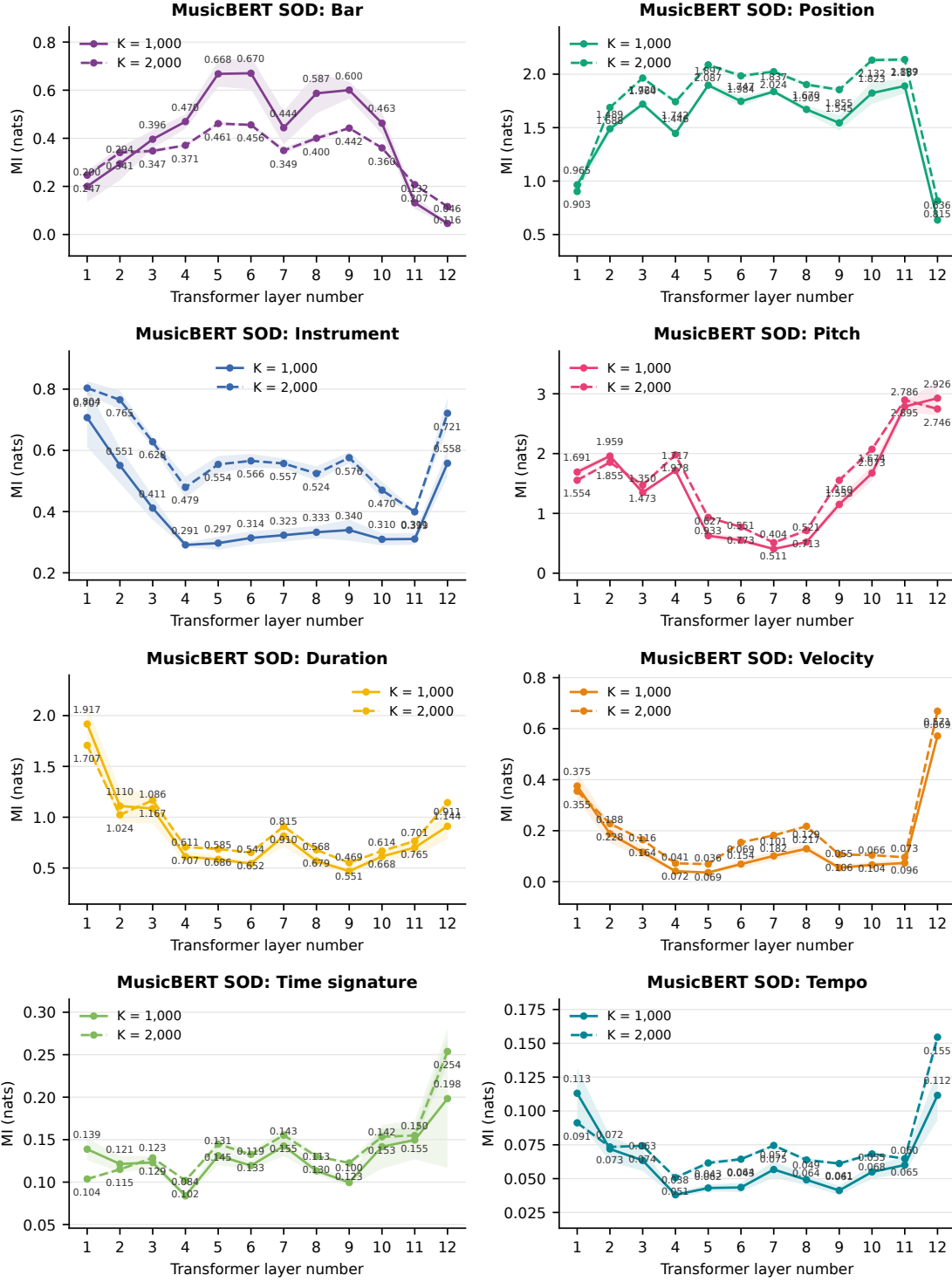


Table 3: SOD: plug-in MI in nats at $K = 1,000$, mean $\pm$ SD over quantizer seeds (12, 42, 97); 4,793,154 note tokens.

Layer	Bar	Position	Instrument	Pitch	Duration	Velocity	Time signature	Tempo
1	0.200 $\pm$ 0.065	0.965 $\pm$ 0.093	0.707 $\pm$ 0.095	1.691 $\pm$ 0.059	1.917 $\pm$ 0.184	0.375 $\pm$ 0.054	0.139 $\pm$ 0.013	0.113 $\pm$ 0.019
2	0.294 $\pm$ 0.069	1.489 $\pm$ 0.042	0.551 $\pm$ 0.059	1.959 $\pm$ 0.037	1.110 $\pm$ 0.164	0.188 $\pm$ 0.037	0.121 $\pm$ 0.010	0.072 $\pm$ 0.009
3	0.396 $\pm$ 0.037	1.720 $\pm$ 0.022	0.411 $\pm$ 0.039	1.350 $\pm$ 0.052	1.086 $\pm$ 0.145	0.116 $\pm$ 0.024	0.123 $\pm$ 0.008	0.063 $\pm$ 0.008
4	0.470 $\pm$ 0.030	1.446 $\pm$ 0.028	0.291 $\pm$ 0.007	1.717 $\pm$ 0.016	0.611 $\pm$ 0.058	0.041 $\pm$ 0.005	0.084 $\pm$ 0.002	0.038 $\pm$ 0.002
5	0.668 $\pm$ 0.054	1.897 $\pm$ 0.025	0.297 $\pm$ 0.022	0.627 $\pm$ 0.052	0.585 $\pm$ 0.029	0.036 $\pm$ 0.005	0.131 $\pm$ 0.011	0.043 $\pm$ 0.003
6	0.670 $\pm$ 0.067	1.747 $\pm$ 0.049	0.314 $\pm$ 0.022	0.551 $\pm$ 0.056	0.544 $\pm$ 0.022	0.069 $\pm$ 0.015	0.119 $\pm$ 0.005	0.043 $\pm$ 0.003
7	0.444 $\pm$ 0.069	1.837 $\pm$ 0.014	0.323 $\pm$ 0.021	0.404 $\pm$ 0.012	0.815 $\pm$ 0.105	0.101 $\pm$ 0.019	0.143 $\pm$ 0.012	0.057 $\pm$ 0.006
8	0.587 $\pm$ 0.084	1.670 $\pm$ 0.036	0.333 $\pm$ 0.020	0.521 $\pm$ 0.010	0.568 $\pm$ 0.050	0.129 $\pm$ 0.016	0.113 $\pm$ 0.005	0.049 $\pm$ 0.004
9	0.600 $\pm$ 0.036	1.545 $\pm$ 0.072	0.340 $\pm$ 0.034	1.150 $\pm$ 0.102	0.469 $\pm$ 0.026	0.055 $\pm$ 0.011	0.100 $\pm$ 0.005	0.041 $\pm$ 0.003
10	0.463 $\pm$ 0.046	1.823 $\pm$ 0.102	0.310 $\pm$ 0.020	1.674 $\pm$ 0.130	0.614 $\pm$ 0.046	0.066 $\pm$ 0.011	0.142 $\pm$ 0.026	0.055 $\pm$ 0.005
11	0.132 $\pm$ 0.025	1.889 $\pm$ 0.069	0.311 $\pm$ 0.019	2.786 $\pm$ 0.096	0.701 $\pm$ 0.086	0.073 $\pm$ 0.013	0.150 $\pm$ 0.023	0.060 $\pm$ 0.006
12	0.046 $\pm$ 0.009	0.636 $\pm$ 0.053	0.558 $\pm$ 0.045	2.926 $\pm$ 0.282	0.911 $\pm$ 0.130	0.571 $\pm$ 0.020	0.198 $\pm$ 0.081	0.112 $\pm$ 0.018

Table 4: SOD: plug-in MI in nats at $K = 2,000$, mean $\pm$ SD over quantizer seeds (1, 2, 3, 4, 42); 15,660,784 note tokens.

Layer	Bar	Position	Instrument	Pitch	Duration	Velocity	Time signature	Tempo
1	0.247 $\pm$ 0.006	0.903 $\pm$ 0.017	0.804 $\pm$ 0.022	1.554 $\pm$ 0.040	1.707 $\pm$ 0.026	0.355 $\pm$ 0.014	0.104 $\pm$ 0.002	0.091 $\pm$ 0.001
2	0.341 $\pm$ 0.004	1.688 $\pm$ 0.015	0.765 $\pm$ 0.030	1.855 $\pm$ 0.011	1.024 $\pm$ 0.016	0.228 $\pm$ 0.007	0.115 $\pm$ 0.002	0.073 $\pm$ 0.001
3	0.347 $\pm$ 0.009	1.964 $\pm$ 0.020	0.628 $\pm$ 0.015	1.473 $\pm$ 0.022	1.167 $\pm$ 0.014	0.164 $\pm$ 0.004	0.129 $\pm$ 0.001	0.074 $\pm$ 0.001
4	0.371 $\pm$ 0.005	1.742 $\pm$ 0.020	0.479 $\pm$ 0.035	1.978 $\pm$ 0.028	0.707 $\pm$ 0.015	0.072 $\pm$ 0.003	0.102 $\pm$ 0.001	0.051 $\pm$ 0.001
5	0.461 $\pm$ 0.007	2.087 $\pm$ 0.016	0.554 $\pm$ 0.028	0.933 $\pm$ 0.016	0.686 $\pm$ 0.010	0.069 $\pm$ 0.003	0.145 $\pm$ 0.004	0.062 $\pm$ 0.002
6	0.456 $\pm$ 0.005	1.984 $\pm$ 0.025	0.566 $\pm$ 0.023	0.773 $\pm$ 0.016	0.652 $\pm$ 0.012	0.154 $\pm$ 0.001	0.133 $\pm$ 0.004	0.064 $\pm$ 0.001
7	0.349 $\pm$ 0.011	2.024 $\pm$ 0.020	0.557 $\pm$ 0.017	0.511 $\pm$ 0.014	0.910 $\pm$ 0.021	0.182 $\pm$ 0.005	0.155 $\pm$ 0.007	0.075 $\pm$ 0.001
8	0.400 $\pm$ 0.013	1.903 $\pm$ 0.020	0.524 $\pm$ 0.023	0.713 $\pm$ 0.032	0.679 $\pm$ 0.009	0.217 $\pm$ 0.005	0.130 $\pm$ 0.002	0.064 $\pm$ 0.002
9	0.442 $\pm$ 0.003	1.855 $\pm$ 0.014	0.576 $\pm$ 0.019	1.553 $\pm$ 0.044	0.551 $\pm$ 0.008	0.106 $\pm$ 0.004	0.123 $\pm$ 0.002	0.061 $\pm$ 0.001
10	0.360 $\pm$ 0.008	2.132 $\pm$ 0.023	0.470 $\pm$ 0.028	2.073 $\pm$ 0.041	0.668 $\pm$ 0.008	0.104 $\pm$ 0.004	0.153 $\pm$ 0.010	0.068 $\pm$ 0.002
11	0.207 $\pm$ 0.004	2.137 $\pm$ 0.021	0.399 $\pm$ 0.005	2.895 $\pm$ 0.035	0.765 $\pm$ 0.015	0.096 $\pm$ 0.002	0.155 $\pm$ 0.008	0.065 $\pm$ 0.001
12	0.116 $\pm$ 0.001	0.815 $\pm$ 0.025	0.721 $\pm$ 0.050	2.746 $\pm$ 0.027	1.144 $\pm$ 0.036	0.669 $\pm$ 0.023	0.254 $\pm$ 0.015	0.155 $\pm$ 0.006